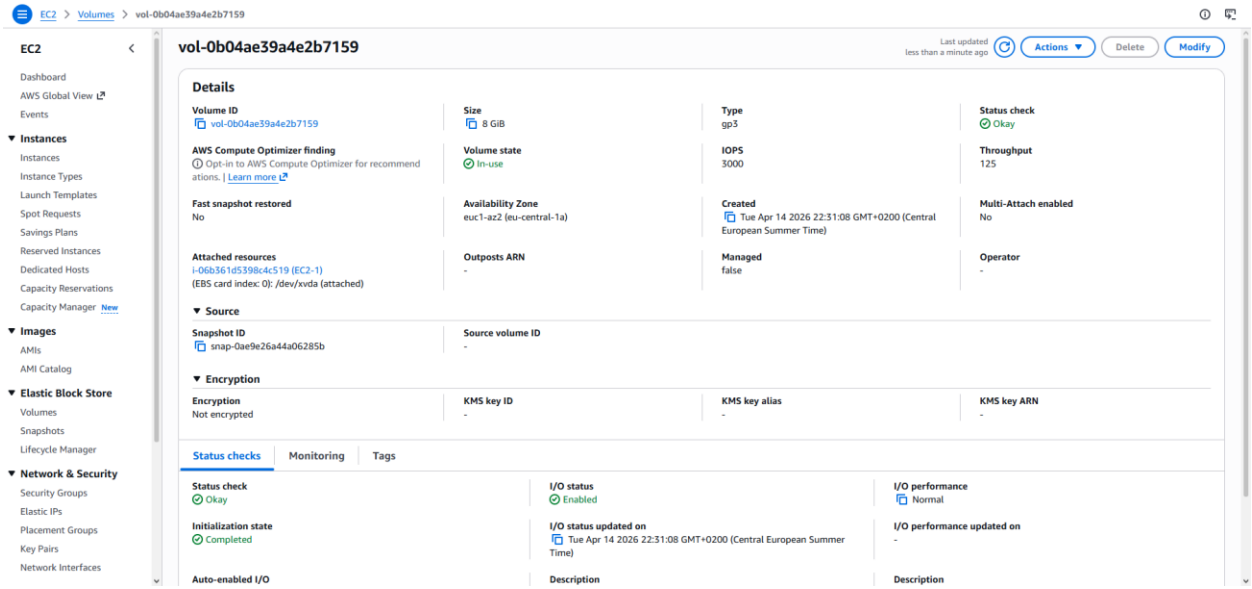
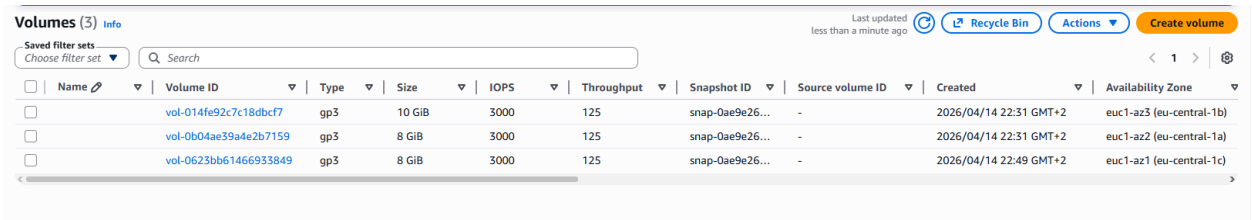
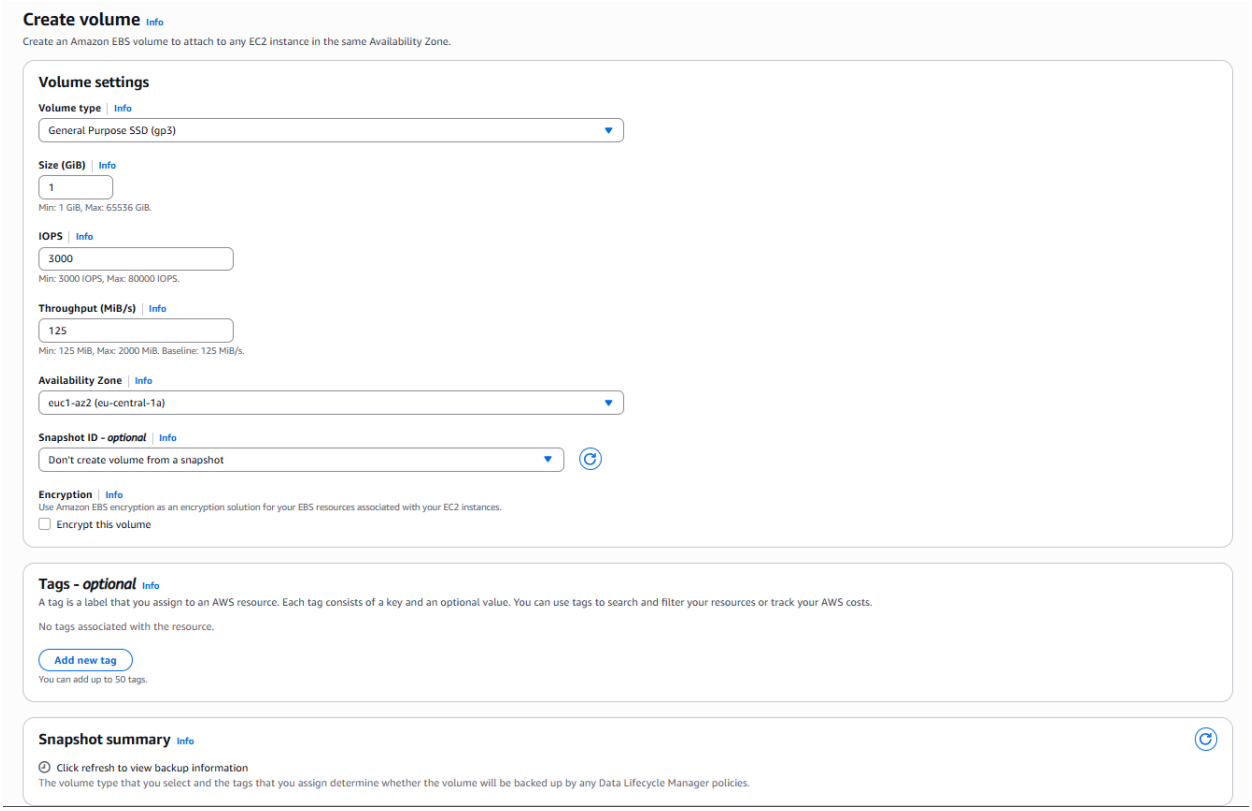
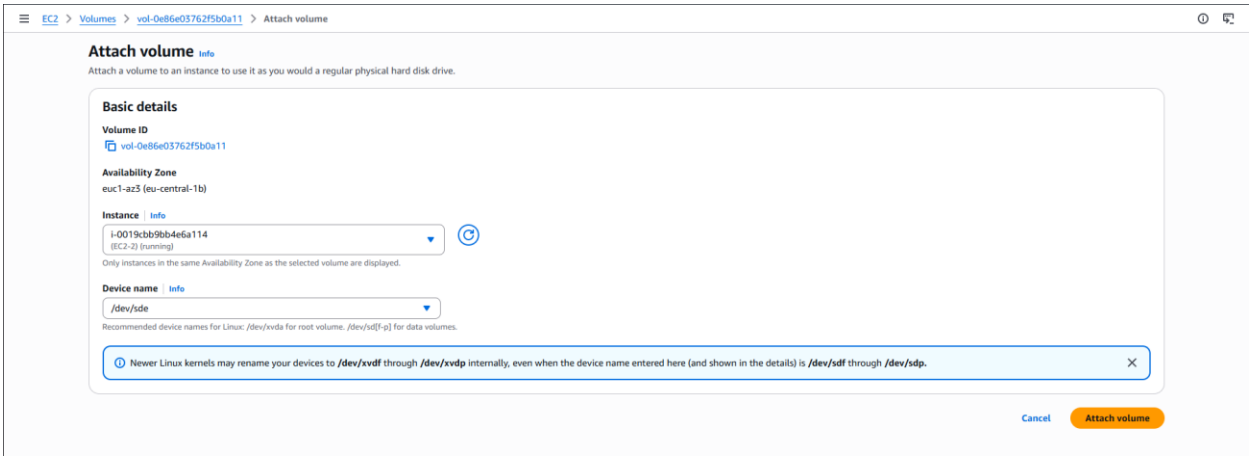
# Domača naloga 8 – Ibrahim Čaušević

Srečanje 8 – delo na predavanjih:






Kreiranje RDS podatkovne baze:

Aurora and RDS > Databases > Create database

# Create database

Free plan has access to limited features and resources

Upgradeable

## Engine options

Engine type

Aurora (MySQL Compatible) | Aurora (PostgreSQL Compatible) | MySQL | PostgreSQL | MariaDB | Oracle | Microsoft SQL Server | IBM Db2

## Choose a database creation method

Full configuration | Easy create

## Templates

Choose a sample template to meet your use case.

Production | Dev/Test | Free tier

## Settings

Engine version

Hide filters

Show only versions that support the Amazon RDS Optimized Writes

Engine version

MySQL 8.0.36

DB instance identifier

database-date-2026

Credentials Settings

Master username

admin

Credentials management

Managed in AWS Secrets Manager - most secure | Self managed

Auto generate password

You can view your credentials after you create your database. Click the View credential details button in the database creation form to view the password.

Additional credentials settings

Database authentication options

Password authentication | Password and IAM database authentication

## Instance configuration

DB instance class

Hide filters

Show previous classes that support Amazon RDS Optimized Writes

Include previous generation classes

Standard classes (includes m classes)

Memory optimized classes (includes r and x classes)

Burstable classes (includes t classes)

Instance type

db.t4g.micro

2 vCPU | 1 GiB RAM | Up to 2,085 Mbps | Network: Up to 5 Gbps

## Storage

Storage type

General Purpose SSD (gp3)

Performance determined by volume size

Allocated storage

20 GiB

Allowed storage values: from 20 GiB to 16,384 GiB.

Additional storage configuration

## Availability & durability

Multi-AZ deployment

Create a standby instance (recommended for production usage)

Do not create a standby instance

## Connectivity

Compute resource

Don't connect to an EC2 compute resource | Connect to an EC2 compute resource

Network type

IPv4 | Dual-stack mode

Virtual private cloud (VPC)

default-vpc (vpc-0f1d1f1f1f1f1f1f1)

3 Subnets, 3 Availability Zones

After a database is created, you can't change its VPC. Only VPCs in the corresponding AWS Region are listed.

Allowed database in another AWS account, you can't change its VPC

DB subnet group

Create new DB Subnet Group

Public access

Yes

No

VPC security group (firewall)

Choose existing | Create new

Existing VPC security groups

default

Availability Zone

No preference

RDS Proxy

Create an RDS Proxy

Certificate authority - optional

rds-ca-rsa2048-g1 (default)

Additional configuration

## Tags - optional

A tag consists of a case-sensitive key-value pair.

No tags associated with this resource.

Add new tag

You can add up to 50 more tags.

## Monitoring

Database Insights - Advanced | Database Insights - Standard

Additional monitoring settings

Enhanced Monitoring

Enable enhanced monitoring

Log exports

Audit log

Error log

General log

Slow query log

IAM role

RDS service-linked role

## Additional configuration

Database port: 3306

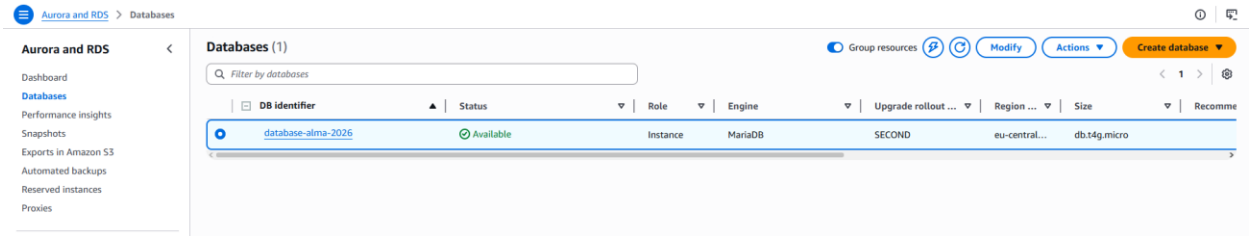
Encryption: Enabled

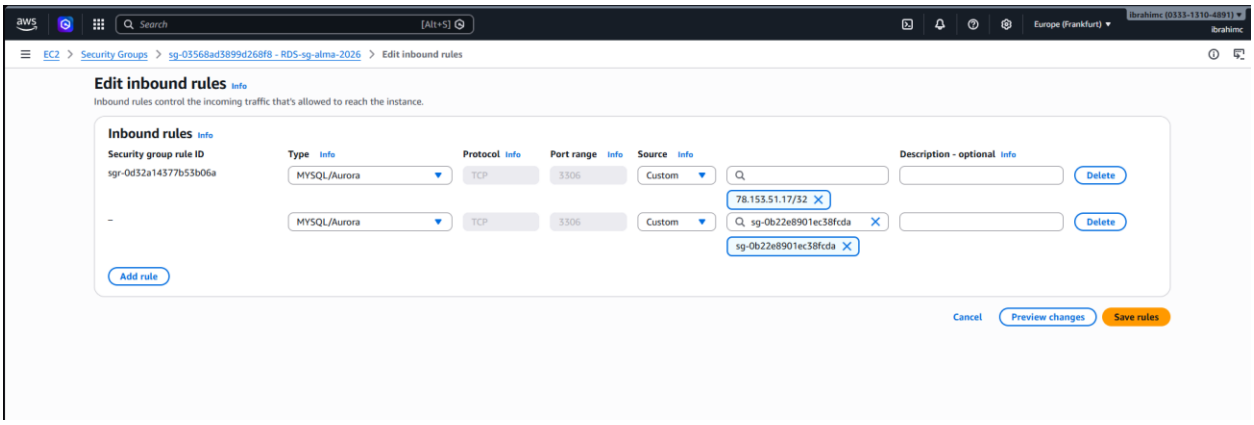
Kerberos authentication: not chosen

Deletion protection: Disabled

You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use in AWS services.

Cancel | Create database




Namestitev mariadb-client na EC2-spletni:

admin@ip-192-168-0-121: ~

```
login as: admin
Authenticating with public key "keyIbrahim"
Linux ip-192-168-0-121 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian
 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Apr 14 20:52:50 2026 from 78.153.51.17
admin@ip-192-168-0-121:~$ sudo apt update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [38 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [47 B]
Hit:3 https://cdn-aws.deb.debian.org/debian trixie InRelease
Get:4 https://cdn-aws.deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:5 https://cdn-aws.deb.debian.org/debian trixie-backports InRelease [54.0 kB]
Get:6 https://cdn-aws.deb.debian.org/debian-security trixie-security InRelease [
43.4 kB]
Get:7 https://cdn-aws.deb.debian.org/debian trixie-backports/main Sources.diff/I
ndex [63.3 kB]
Get:8 https://cdn-aws.deb.debian.org/debian trixie-backports/main amd64 Packages
.diff/Index [63.3 kB]
Get:12 https://cdn-aws.deb.debian.org/debian trixie-backports/main Sources T-202
6-04-21-0802.56-F-2026-04-20-1400.07.pdiff [1306 B]
Get:12 https://cdn-aws.deb.debian.org/debian trixie-backports/main Sources T-202
6-04-21-0802.56-F-2026-04-20-1400.07.pdiff [1306 B]
Get:13 https://cdn-aws.deb.debian.org/debian trixie-backports/main amd64 Package
s T-2026-04-21-0802.56-F-2026-04-20-2000.10.pdiff [497 B]
Get:13 https://cdn-aws.deb.debian.org/debian trixie-backports/main amd64 Package
s T-2026-04-21-0802.56-F-2026-04-20-2000.10.pdiff [497 B]
Get:9 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Source
s [130 kB]
Get:10 https://cdn-aws.deb.debian.org/debian-security trixie-security/main amd64
 Packages [125 kB]
Get:11 https://cdn-aws.deb.debian.org/debian-security trixie-security/main Trans
lation-en [78.8 kB]
Fetched 607 kB in 0s (2032 kB/s)
All packages are up to date.
admin@ip-192-168-0-121:~$ sudo apt upgrade
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
admin@ip-192-168-0-121:~$ sudp apt install mariadb-client
-bash: sudp: command not found
admin@ip-192-168-0-121:~$ sudo apt install mariadb-client
Installing:
  mariadb-client

Installing dependencies:
  libconfig-inifiles-perl  libncurses6          mariadb-common
  libdbd-mariadb-perl      libpcre2-posix3      mysql-common
  libdbi-perl              libterm-readkey-perl
  libmariadb3              mariadb-client-core

Suggested packages:
  libclone-perl  libmldbm-perl  libnet-daemon-perl  libsql-statement-perl

Summary:
  Upgrading: 0, Installing: 11, Removing: 0, Not Upgrading: 0
  Download size: 5498 kB
  Space needed: 85.9 MB / 6630 MB available
```

```
admin@ip-192-168-0-121:~$ mariadb -h database-alma-2026.cteqq0ge0wbv.eu-central-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 79
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Kreiranje PB in USER-ja v novi PB:

admin@ip-192-168-0-121: /var/www/html

```
admin@ip-192-168-0-121:/var/www$ cd /html
-bash: cd: /html: No such file or directory
admin@ip-192-168-0-121:/var/www$ cd /var/www/html
admin@ip-192-168-0-121:/var/www/html$ sudo nano vstavi.php
admin@ip-192-168-0-121:/var/www/html$ sudo nano vstavi.php
admin@ip-192-168-0-121:/var/www/html$ sudo nano izpis.php
admin@ip-192-168-0-121:/var/www/html$ mariadb -h database-alma-2026.cteqq0ge0wbv.eu-central-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 113
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.004 sec)

MariaDB [(none)]> CREATE DATABASE nakup;
Query OK, 1 row affected (0.008 sec)

MariaDB [(none)]> USE nakup;
Database changed
MariaDB [nakup]> CREATE TABLE nakup (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     element VARCHAR(100) NOT NULL,
    ->     kolicina INT NOT NULL
    -> );
Query OK, 0 rows affected (0.031 sec)

MariaDB [nakup]> CREATE USER 'ibrahim'@'%' IDENTIFIED BY 'ibrahim';
Query OK, 0 rows affected (0.021 sec)

MariaDB [nakup]> GRANT ALL ON  *.* TO 'ibrahim'@'%';
ERROR 1045 (28000): Access denied for user 'admin'@'%' (using password: YES)
MariaDB [nakup]> GRANT ALL ON  *.* TO 'ibrahim'@'%';
ERROR 1045 (28000): Access denied for user 'admin'@'%' (using password: YES)
MariaDB [nakup]> GRANT ALL ON  nakup.* TO 'ibrahim'@'%';
Query OK, 0 rows affected (0.007 sec)
```

Nastavljanje PHP aplikacije z novimi nastavitvami:
VSTAVI.PHP

admin@ip-192-168-0-121: /var/www/html

```
  GNU nano 8.4
<?php
error_reporting(E_ALL);
ini_set('display_errors', 1);

mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);

$host = "database-alma-2026.cteqq0ge0wbv.eu-central-1.rds.amazonaws.com";
$port = 3306;
$dbname = "nakup";
$username = "ibrahim";
$password = "ibrahim";
$ca = "/etc/ssl/rds/global-bundle.pem";

$element = trim($_POST['element'] ?? '');
$kolicina = (int)($_POST['kolicina'] ?? 0);

if ($element === '' || $kolicina <= 0) {
    die("Element in veljavna količina sta obvezna.");
}
try {
    $conn = mysqli_init();
    $conn->ssl_set(NULL, NULL, $ca, NULL, NULL);
    $conn->real_connect($host, $username, $password, $dbname, $port, NULL, MYSQLI_CLIENT_SSL);

    $conn->set_charset("utf8mb4");

    $stmt = $conn->prepare("INSERT INTO nakup (element, kolicina) VALUES (?, ?)");
    $stmt->bind_param("si", $element, $kolicina);
    $stmt->execute();

    echo "Podatek je bil uspešno shranjen.<br><br>";
    echo '<a href="index.html">Nazaj na vnos</a><br>';
    echo '<a href="izpis.php">Poglej seznam</a>';

    $stmt->close();
    $conn->close();
} catch (mysqli_sql_exception $e) {
    die("MySQL napaka: " . $e->getMessage());
}
?>
```

IZPIS.PHP

```
  GNU nano 8.4                                                                    izpis.php
<?php
ini_set('display_errors', 1);
error_reporting(E_ALL);

$host = "database-alma-2026.cteqq0ge0wbv.eu-central-1.rds.amazonaws.com";
$dbname = "nakup";
$username = "ibrahim";
$password = "ibrahim";
$ca = "/etc/ssl/rds/global-bundle.pem";

try {
    mysqli_report(MYSQLI_REPORT_ERROR | MYSQLI_REPORT_STRICT);

    $conn = mysqli_init();
    $conn->ssl_set(NULL, NULL, $ca, NULL, NULL);
    $conn->real_connect($host, $username, $password, $dbname, 3306, NULL, MYSQLI_CLIENT_SSL);

    $conn->set_charset("utf8mb4");

    $sql = "SELECT id, element, kolicina FROM nakup";
    $result = $conn->query($sql);

    echo "<h1>Nakupni seznam</h1>";

    if ($result && $result->num_rows > 0) {
        while ($row = $result->fetch_assoc()) {
            echo "ID: " . htmlspecialchars($row["id"]) .
                 " - Element: " . htmlspecialchars($row["element"]) .
                 " - Količina: " . htmlspecialchars($row["kolicina"]) . "<br>";
        }
    } else {
        echo "Ni podatkov.";
    }

    echo "<br><a href='index.html'>Nazaj</a>";

    $conn->close();

} catch (mysqli_sql_exception $e) {
    die("MySQL napaka: " . $e->getMessage());
}
?>
```

Prikaz IZPIS.PHP v brskalniku:

# Nakupni seznam

ID: 1 - Element: Čokolada - Količina: 3
ID: 2 - Element: Jajca - Količina: 5
ID: 3 - Element: Mleko - Količina: 3

Nazaj